

COS1501

UNIT 1

The Development of Number Systems

$$\mathbb{Z}^+ \bullet \mathbb{Z}_{\geq 0} \bullet \mathbb{Z}$$

Complete, guide-aligned notes at the original academic level

Every assessed topic includes a medium and a really hard fully worked example

Colour key

TEAL
definitions and laws

BLUE
medium worked
examples

PURPLE
really hard worked
examples

RED / ORANGE
traps and exam notes

Unit roadmap and scope

This document follows Study Unit 1 of the supplied COS1501 guide. It preserves the guide's notation and formal statements while adding deeper explanations, property-labelled algebra, counterexamples, and harder applications.

EXAM NOTE - Required standard

An example can illustrate a universal law, but finitely many examples cannot prove that the law holds for every integer. One counterexample, however, is enough to disprove a universal claim.

PART	FOCUS	COMPLETE COVERAGE
1	Number systems and notation	$\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$ (or $\mathbb{N}$), $\mathbb{Z}$; membership; containment; closure
2	Laws on $\mathbb{Z}^+$	commutativity, associativity, distributivity, multiplicative identity
3	The extension to $\mathbb{Z}_{\geq 0}$	zero, place value, additive identity, zero-product law
4	The extension to $\mathbb{Z}$	negative integers, number line, addition, multiplication signs
5	Core integer concepts	additive inverse, equations, order, absolute value, primes, factorial
6	Formal law system	the nine laws for $\mathbb{Z}_{\geq 0}$; changed monotonicity and Law 10 for $\mathbb{Z}$

Key questions you must be able to answer

- How do $\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$ differ, and which operations are closed in each system?
- Why did zero make positional notation and algebraic identities possible?
- Why do the four integer multiplication sign rules follow from the ordinary laws?
- How are identity, inverse, absolute value, primality and factorial defined formally?
- Which of the nine laws hold in $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$, and how does multiplication affect inequalities?

1. The three number systems: $\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$

DEFINITION - Sets used in Unit 1

Positive integers: $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.

Non-negative integers: $\mathbb{Z}_{\geq 0} = \{0, 1, 2, 3, \dots\}$. In this guide, $\mathbb{N}$ means this set.

Integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

The containment chain is $\mathbb{Z}^+ \subset \mathbb{Z}_{\geq 0} \subset \mathbb{Z}$. Every positive integer is non-negative, and every non-negative integer is an integer. The converse statements fail: 0 is not positive, and -1 is not non-negative.

A system is closed under an operation when applying that operation to members of the system always returns a member of the same system. All three systems are closed under addition and multiplication. $\mathbb{Z}$ is closed under subtraction, but $\mathbb{Z}^+$ and $\mathbb{Z}_{\geq 0}$ are not: $2 - 5 = -3$ lies outside both.

COMMON TRAP - Notation warning

Some books define $\mathbb{N} = \{1, 2, 3, \dots\}$. COS1501 Unit 1 uses $\mathbb{N}$ for $\{0, 1, 2, 3, \dots\}$. State the convention before using $\mathbb{N}$ in formal work.

MEDIUM WORKED EXAMPLE - Classify numbers and operations

Problem. Classify -7, 0 and 12 in the three systems, then decide whether $4 - 9$ is closed in $\mathbb{Z}_{\geq 0}$.

Solution.

$-7 \in \mathbb{Z}$ only. $0 \in \mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$, but $0 \notin \mathbb{Z}^+$. $12 \in \mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$.

Because $4, 9 \in \mathbb{Z}_{\geq 0}$ but $4 - 9 = -5 \notin \mathbb{Z}_{\geq 0}$, subtraction is not closed in $\mathbb{Z}_{\geq 0}$.

REALLY HARD WORKED EXAMPLE - Determine the exact system

Problem. Let A be the smallest one of $\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$, $\mathbb{Z}$ containing every value of $a - b$, where $a, b \in \mathbb{Z}^+$. Identify A and justify why each smaller candidate fails.

Solution.

Positive integers can produce positive, zero or negative differences: $8 - 3 = 5$, $4 - 4 = 0$, and $2 - 9 = -7$. Therefore all possible values are integers, so $A = \mathbb{Z}$. $\mathbb{Z}^+$ fails because it excludes 0 and negatives; $\mathbb{Z}_{\geq 0}$ fails because it excludes negative differences.

2. Variables, quantifiers and multiplication notation

Letters such as m , n and k are variables. A statement beginning 'for all positive integers m and n ' represents infinitely many specific statements at once. The domain restriction is part of the statement; ignoring it changes the claim.

Products may be written as $(m)(n)$, mn or $m \cdot n$. The guide avoids $m \times n$ because $\times$ may be confused with the variable x and later represents the Cartesian product of sets. With numerical substitutions, use a product sign or brackets: $(113)(25)$, not 11325 .

EXAM NOTE - Logical precision

Substitution is a check or illustration, not a proof of a universal statement. To disprove a universal statement, search for a counterexample that satisfies the stated domain.

MEDIUM WORKED EXAMPLE - Read a quantified law

Problem. Expand 'for all $m, n \in \mathbb{Z}^+$, $mn = nm$ ' using $m=113$ and $n=25$, with unambiguous notation.

Solution.

The instance is $(113)(25) = (25)(113)$. Both sides equal 2825. Writing 11325 would mean the integer eleven-thousand three-hundred twenty-five, not a product.

REALLY HARD WORKED EXAMPLE - Find a legal counterexample

Problem. A student claims: for all $m, n \in \mathbb{Z}$, $m + n = mn$. Disprove the claim, and explain why choosing $m=1/2$ would be a poor counterexample in this context.

Solution.

Choose $m=2$ and $n=3$, both in $\mathbb{Z}$. Then $m+n=5$ while $mn=6$, so the universal claim is false. The value $1/2$ is not in the declared domain $\mathbb{Z}$; even if it produced unequal sides, it would not be a legal counterexample to this statement.

3. Commutative property

DEFINITION - Formal law

For all m, n in the stated integer system:

Addition: $m+n = n+m$.

Multiplication: $mn = nm$.

Commutativity changes the order of two operands, not their grouping.

Subtraction is not commutative: in general $m-n \neq n-m$. Division is also not commutative. The law belongs to a particular operation, not automatically to every operation on the set.

MEDIUM WORKED EXAMPLE - Reorder for efficient calculation

Problem. Evaluate $47 + 125 + 53 + 75$, explicitly using commutativity.

Solution.

Reorder the terms: $47+125+53+75 = 47+53+125+75 = 100+200 = 300$. The swaps use commutativity; the regrouping into pairs uses associativity.

REALLY HARD WORKED EXAMPLE - Property-labelled rearrangement

Problem. Show that $(ab)(cd) = (ac)(bd)$ for positive integers a, b, c, d using only commutativity and associativity.

Solution.

$(ab)(cd) = ((ab)c)d$ by associativity

$= ((ac)b)d$ by commutativity applied to b and c within the product

$= (ac)(bd)$ by associativity.

The equality is not a new law; it is a consequence of the two named laws.

4. Associative property

DEFINITION - Formal law

For all m, n, k in the stated integer system:

Addition: $m + (n + k) = (m + n) + k$.

Multiplication: $m(nk) = (mn)k$.

Associativity changes grouping, while preserving order.

Because repeated additions and multiplications may be regrouped without changing the value, long strings such as $a + b + c + d$ can be written without brackets. Subtraction fails associativity: $10 - (6 - 2) = 6$, but $(10 - 6) - 2 = 2$.

MEDIUM WORKED EXAMPLE - Group convenient totals

Problem. Evaluate $38 + (62 + 145)$ without changing the order of the terms.

Solution.

$38 + (62 + 145) = (38 + 62) + 145$ by associativity $= 100 + 145 = 245$.

REALLY HARD WORKED EXAMPLE - Reverse a fully nested sum

Problem. Transform $((7 + 13) + 5) + 9$ into $((9 + 5) + 13) + 7$, naming the property types used.

Solution.

First commute whole or inner addends to obtain $9 + ((7 + 13) + 5)$, then commute within groups to obtain $9 + (5 + (13 + 7))$. Use associativity twice: $9 + ((5 + 13) + 7) = (9 + (5 + 13)) + 7 = ((9 + 5) + 13) + 7$. Commutativity changes order; associativity changes brackets.

5. Distributive property

DEFINITION - Formal law

For all m, n, k in the stated integer system, multiplication distributes over addition:

$$m(n+k) = mn+mk.$$

By commutativity, $(n+k)m = nm+km$.

Read the law left-to-right to expand; read it right-to-left to factor out a common factor. It extends to more than two terms: $a(b+c+d)=ab+ac+ad$.

COMMON TRAP - False reverse idea

Addition is not distributive over multiplication. For $m=2, n=3, k=4$: $m+(nk)=14$, while $(m+n)(m+k)=30$. One counterexample destroys the universal claim.

MEDIUM WORKED EXAMPLE - Expand and collect

Problem. Simplify $(3x+2)(x+4)$ using distributivity at every step.

Solution.

$$\begin{aligned}(3x+2)(x+4) &= (3x+2)x + (3x+2)4 \\ &= 3x^2+2x+12x+8 = 3x^2+14x+8.\end{aligned}$$

REALLY HARD WORKED EXAMPLE - Factor completely

Problem. Factor $42x^6 + 63x^5 - 105x^4$ by repeated use of distributivity.

Solution.

The greatest common factor is $21x^4$:

$$42x^6+63x^5-105x^4 = 21x^4(2x^2+3x-5).$$

Now $2x^2+3x-5 = (2x+5)(x-1)$, because $2x^2-2x+5x-5 = 2x^2+3x-5$.

Therefore the complete factorisation is $21x^4(2x+5)(x-1)$.

6. Multiplicative identity

DEFINITION - Formal law

There exists an element 1 such that, for every m , $m \cdot 1 = m$. By commutativity, $1 \cdot m = m$. The element leaves every number unchanged under multiplication.

Identity is operation-specific: 1 is the identity for multiplication, but not for addition. Writing a carefully chosen form of 1 is a standard algebraic technique because it changes appearance without changing value.

MEDIUM WORKED EXAMPLE - Insert the identity

Problem. Show why $37 \cdot 1 \cdot (5+2)$ has the same value as $37(5+2)$, and evaluate it.

Solution.

$37 \cdot 1 = 37$ by multiplicative identity, so $37 \cdot 1 \cdot (5+2) = 37 \cdot 7 = 259$.

REALLY HARD WORKED EXAMPLE - Prove uniqueness of the multiplicative identity

Problem. Suppose e and f both act as multiplicative identities in $\mathbb{Z}^+$. Prove $e=f$.

Solution.

Because f is an identity, $e \cdot f = e$. Because e is an identity, $e \cdot f = f$. The same product equals both e and f , hence $e=f$. Therefore two distinct multiplicative identities cannot exist.

7. Extending $\mathbb{Z}^+$ to $\mathbb{Z}_{\geq 0}$: zero is a number

$\mathbb{Z}_{\geq 0}$ adds exactly one element to $\mathbb{Z}^+$: zero. Zero is not the same as absence or missing data. A mark of 0 is an assigned numerical result; no mark means no numerical result exists.

Adding zero preserves commutativity, associativity, distributivity and multiplicative identity. It also introduces the additive identity and enables a complete place-value notation with placeholders.

NOTE - Set relation

$\mathbb{Z}_{\geq 0} = \mathbb{Z}^+ \cup \{0\}$. Consequently $\mathbb{Z}^+ \subset \mathbb{Z}_{\geq 0}$, but $0 \notin \mathbb{Z}^+$.

MEDIUM WORKED EXAMPLE - Zero versus missing information

Problem. A database stores 0 for one student's mark and NULL for another. Are the records mathematically equivalent?

Solution.

No. The value 0 is a member of $\mathbb{Z}_{\geq 0}$ and can enter arithmetic: $0+8=8$. NULL indicates that no value is recorded, so ordinary integer arithmetic is not defined on it.

REALLY HARD WORKED EXAMPLE - Identify the minimal extension

Problem. Starting from $\mathbb{Z}^+$, add as few elements as possible so that addition has an identity within the system. What must be added, and why can no positive integer serve?

Solution.

An additive identity e must satisfy $m+e=m$ for every positive m . Taking $m=1$ gives $1+e=1$, forcing $e=0$. No positive integer works because $1+e>1$ for every $e \in \mathbb{Z}^+$. Thus the minimal required extension adds 0, producing $\mathbb{Z}_{\geq 0}$.

8. Positional notation, base and place value

Choose a base $b > 1$. The allowed digits are $0, 1, \dots, b-1$. In a numeral $d_n \dots d_2 d_1 d_0$, the rightmost position has value b^0 , the next b^1 , then b^2 , and so on:

$$(d_n \dots d_2 d_1 d_0)_b = d_n b^n + \dots + d_2 b^2 + d_1 b + d_0$$

Zero is a placeholder: 305 means $3 \cdot 10^2 + 0 \cdot 10^1 + 5 \cdot 10^0$. Without the internal zero, 35 would represent a different number. Appending a zero in base b multiplies the represented number by b .

COMMON TRAP - Digit restriction

A digit must be smaller than the base. For example, 7 is not a valid digit in base 7, and 2 is not a valid binary digit.

MEDIUM WORKED EXAMPLE - Evaluate a base-seven numeral

Problem. Convert $(3052)_7$ to decimal.

Solution.

$$(3052)_7 = 3 \cdot 7^3 + 0 \cdot 7^2 + 5 \cdot 7 + 2 = 3 \cdot 343 + 35 + 2 = 1066.$$

REALLY HARD WORKED EXAMPLE - Recover a missing digit

Problem. The base-six numeral $(4a03)_6$ equals 915 in decimal. Determine the digit a and verify it is legal.

Solution.

$$(4a03)_6 = 4 \cdot 6^3 + a \cdot 6^2 + 0 \cdot 6 + 3 = 864 + 36a + 3 = 867 + 36a.$$

Set $867 + 36a = 915$, so $36a = 48$. This gives $a = 4/3$, which is not an integer digit. Therefore no legal digit exists: the information is inconsistent. A hard place-value question may require detecting impossibility, not forcing an answer.

NOTE - Correction insight

If the decimal value had been 1011, then $867 + 36a = 1011$ would give $a = 4$, a legal base-six digit. Always validate the digit range after solving.

9. Additive identity

DEFINITION - Formal law

There exists an element 0 such that, for every non-negative integer or integer m , $m+0=m$. By commutativity, $0+m=m$.

The additive identity is the neutral element for addition. It becomes powerful in equation solving because additive inverses create zero, after which the identity law removes it.

MEDIUM WORKED EXAMPLE - Expose the identity in an equation

Problem. Solve $x+17=17$ using additive inverses and the additive identity.

Solution.

Add -17 to both sides: $x+17+(-17)=17+(-17)$. By associativity, $x+[17+(-17)]=0$, so $x+0=0$, hence $x=0$.

REALLY HARD WORKED EXAMPLE - Prove uniqueness of the additive identity

Problem. Suppose e and f both satisfy $m+e=m$ and $m+f=m$ for every $m \in \mathbb{Z}$. Prove $e=f$.

Solution.

Since f is an additive identity, $e+f=e$. Since e is an additive identity, $e+f=f$. Therefore $e=f$. The additive identity is unique.

10. Multiplication by zero and absence of zero-divisors

DEFINITION - Two-part law

For every integer m , $m \cdot 0 = 0$. Conversely, if $mn = 0$, then $m = 0$ or $n = 0$ (possibly both). The combined statement is $mn = 0$ if and only if $m = 0$ or $n = 0$.

The converse is the zero-product property. It permits a factored equation to split into separate cases. You may not use it while the equation equals a non-zero number.

MEDIUM WORKED EXAMPLE - Solve a factored quadratic

Problem. Solve $(x-6)(x+4)=0$ over $\mathbb{Z}$.

Solution.

By the zero-product property, $x-6=0$ or $x+4=0$. Therefore $x=6$ or $x=-4$. Check: either factor becomes zero.

REALLY HARD WORKED EXAMPLE - Use substitution and repeated zero products

Problem. Solve $x^4 - 13x^2 + 36 = 0$ over $\mathbb{Z}$.

Solution.

Let $u = x^2$. Then $u^2 - 13u + 36 = (u-4)(u-9) = 0$, so $u=4$ or $u=9$.

Thus $x^2=4$ or $x^2=9$, giving $x=\pm 2$ or $x=\pm 3$. The full solution set is $\{-3, -2, 2, 3\}$.

COMMON TRAP - Illegal cancellation

From $ab=ac$ you cannot automatically cancel a unless $a \neq 0$. If $a=0$, both sides equal zero for every b and c . The zero laws explain why cancellation needs a condition.

11. Extending to all integers: $\mathbb{Z}$ and the number line

$\mathbb{Z}$ contains zero, all positive integers, and their negative counterparts. A number line fixes equal spacing and an orientation: values increase to the right and decrease to the left.



Negative integers permit a central reference point with two directions: profit/loss, above/below zero, future/past, increase/decrease. The sign communicates direction; the magnitude communicates distance from zero.

MEDIUM WORKED EXAMPLE - Translate a change

Problem. A balance changes by -R125, then +R80. Represent and determine the net change.

Solution.

The net change is $-125+80=-45$. The final balance is R45 below the starting balance.

REALLY HARD WORKED EXAMPLE - Reconstruct a starting position

Problem. An integer position is moved by -17, then +9, then -6, and ends at -21. Find the starting integer s .

Solution.

$s+(-17)+9+(-6)=-21$. Combine the changes: $-17+9-6=-14$, so $s-14=-21$. Add 14 to both sides: $s=-7$. Check: $-7-17+9-6=-21$.

12. Addition of integers

On the number line, begin at x and move according to y . A positive y moves right; a negative y moves left. The endpoint is $x+y$. Algebraically, adding a negative is the same as moving in the negative direction.

Integer addition remains commutative and associative. These laws let us reorder and regroup positive and negative terms strategically.

MEDIUM WORKED EXAMPLE - Strategic integer sum

Problem. Evaluate $-38+125-62-25$ using named properties.

Solution.

Reorder and regroup: $(-38-62)+(125-25) = -100+100=0$. Reordering uses commutativity; regrouping uses associativity.

REALLY HARD WORKED EXAMPLE - A nested sum with variables

Problem. Given $a+b=-17$ and $b+c=9$, evaluate $(a+b)+(c+(-b))$ without solving separately for a, b, c .

Solution.

Use associativity and commutativity:

$$(a+b)+(c-b) = a+c+(b-b) = a+c.$$

Also $(a+b)+(b+c)=a+c+2b$, so direct use of the given totals does not isolate $a+c$. From the two equations, $a=-17-b$ and $c=9-b$, hence $a+c=-8-2b$. Therefore the expression is $-8-2b$, not a fixed integer. The data are insufficient for one numerical answer.

EXAM NOTE - Hard-question discipline

Do not assume every problem has a unique number. If the givens leave a free variable, the correct conclusion is an expression or 'not uniquely determined'.

13. Multiplication of integers and the sign rules

DEFINITION - Sign rules

$$(+) (+) = + \quad (+) (-) = - \quad (-) (+) = - \quad (-) (-) = +.$$

First multiply magnitudes; then determine the sign from the number of negative factors. An even number of negative factors gives a positive product; an odd number gives a negative product.

For a negative integer x there is a positive a with $x = -a$. The sign rules are not arbitrary. They are forced by distributivity, identity and additive inverses.

MEDIUM WORKED EXAMPLE - Product with many signs

Problem. Evaluate $(-2)(-3)(4)(-5)(-1)$.

Solution.

There are four negative factors, an even number, so the result is positive. The magnitude is $2 \cdot 3 \cdot 4 \cdot 5 \cdot 1 = 120$. Therefore the product is 120.

REALLY HARD WORKED EXAMPLE - Derive minus times minus

Problem. Using distributivity and additive inverses, prove $(-a)(-b) = ab$ for positive integers a, b .

Solution.

Because $b + (-b) = 0$, multiply by $-a$:

$$(-a)[b + (-b)] = (-a) \cdot 0 = 0.$$

By distributivity, $(-a)b + (-a)(-b) = 0$. Since $(-a)b = -(ab)$, we have $-(ab) + (-a)(-b) = 0$. Therefore $(-a)(-b)$ is the additive inverse of $-(ab)$, namely ab . Hence $(-a)(-b) = ab$.

COMMON TRAP - Circular reasoning

Do not use commutativity of integer multiplication to prove a sign case if commutativity itself still depends on those sign cases. A proof cannot assume the result it is meant to establish.

14. Additive inverse

DEFINITION - Formal definition

For every integer x there exists an integer $-x$ such that $x+(-x)=0$. The symbol $-x$ means 'the additive inverse of x '; it does not automatically mean a negative number.

If $x=-2$, then $-x=-(-2)=2$. Subtraction can be defined through addition: $a-b$ means $a+(-b)$.

MEDIUM WORKED EXAMPLE - Solve using an inverse

Problem. Solve $x+23=-11$, displaying the identity step.

Solution.

Add -23 to both sides: $x+23+(-23)=-11+(-23)$. Then $x+0=-34$, so $x=-34$.

REALLY HARD WORKED EXAMPLE - Simplify nested inverses

Problem. Simplify $-[-(a+(-b))]$ and justify the result using the uniqueness of additive inverses.

Solution.

The inner expression is $a-b$. Its additive inverse is $-(a-b)=-a+b$. Taking the additive inverse again returns $a-b$ because the inverse of $-y$ is y : $y+(-y)=0$ and uniqueness identifies y as the inverse of $-y$. Therefore $-[-(a+(-b))]=a-b$.

15. Applying identities and inverses to equations

Unit 1 applies additive identity, additive inverses and the zero-product property to quadratic equations. Two central techniques are factorisation and completing the square.

$$x^2 + bx = c \rightarrow x^2 + bx + (b/2)^2 = c + (b/2)^2 \rightarrow (x+b/2)^2 = \dots$$

The guide's key square-root correction is $\sqrt{(w^2)}=|w|$, not w in every case. When $u^2=v^2$, the valid conclusion is $|u|=|v|$, equivalently $u=v$ or $u=-v$.

MEDIUM WORKED EXAMPLE - Complete the square

Problem. Solve $x^2-6x=7$ over $\mathbb{Z}$.

Solution.

Add 9 to both sides: $x^2-6x+9=16$. Thus $(x-3)^2=16$, so $x-3=4$ or $x-3=-4$. Therefore $x=7$ or $x=-1$.

REALLY HARD WORKED EXAMPLE - Repair a false square-root step

Problem. A student moves from $(2x-5)^2=(x+4)^2$ to $2x-5=x+4$ only. Find every integer solution and identify the lost branch.

Solution.

Equal squares give two cases:

1) $2x-5=x+4$, so $x=9$.

2) $2x-5=-(x+4)$, so $2x-5=-x-4$, hence $3x=1$ and $x=1/3$.

Over $\mathbb{Z}$, only $x=9$ is allowed. Over a wider number system, the second branch also matters. The student's reasoning was invalid even though the integer answer happened to survive.

COMMON TRAP - Square-root rule

The principal square root symbol $\sqrt{q}$ is non-negative. For any integer w , $\sqrt{(w^2)}=|w|$. Dropping the absolute value is valid only when $w \geq 0$ is already known.

16. Ordering integers and the number line

For integers x and y , $x < y$ exactly when x lies to the left of y on the number line. An equivalent test is $x < y$ iff $y - x$ is positive, that is, $y - x \in \mathbb{Z}^+$.



Negative numbers reverse an everyday magnitude intuition: $-2 > -7$ because -2 lies farther right, even though 7 has the larger absolute magnitude.

MEDIUM WORKED EXAMPLE - Compare using a positive difference

Problem. Show $-13 < 5$ without relying only on the picture.

Solution.

Compute $5 - (-13) = 5 + 13 = 18$. Since $18 \in \mathbb{Z}^+$, the definition gives $-13 < 5$.

REALLY HARD WORKED EXAMPLE - Order three algebraic integers

Problem. Let $a = -8$. Arrange a , $-a - 3$ and $a + 11$ in increasing order, and justify with differences.

Solution.

$a = -8$, $-a - 3 = 8 - 3 = 5$, and $a + 11 = 3$. Thus $-8 < 3 < 5$. Differences are $3 - (-8) = 11 > 0$ and $5 - 3 = 2 > 0$, confirming the order.

17. Absolute value

DEFINITION - Piecewise definition

For $x \in \mathbb{Z}$:

$$|x| = x \text{ if } x \geq 0;$$

$$|x| = -x \text{ if } x < 0.$$

It measures distance from zero, so $|x| \geq 0$ and $|x| = 0$ exactly when $x = 0$.

Important consequences: $|-x| = |x|$; $|xy| = |x||y|$; and $|x-y|$ is the distance between x and y . Absolute value compares magnitude, not ordinary order.

MEDIUM WORKED EXAMPLE - Solve a distance equation

Problem. Solve $|x-4|=7$ over $\mathbb{Z}$.

Solution.

The distance from x to 4 is 7, so $x-4=7$ or $x-4=-7$. Therefore $x=11$ or $x=-3$.

REALLY HARD WORKED EXAMPLE - Nested absolute values

Problem. Solve $||x-2|-5|=1$ over $\mathbb{Z}$.

Solution.

Let $u=|x-2|$, where $u \geq 0$. Then $|u-5|=1$, so $u-5=1$ or $u-5=-1$; hence $u=6$ or $u=4$.

If $|x-2|=6$, $x=8$ or -4 . If $|x-2|=4$, $x=6$ or -2 .

Solution set: $\{-4, -2, 6, 8\}$.

COMMON TRAP - Order trap

From $x < y$ you cannot conclude $|x| < |y|$. Example: $-10 < 3$ but $|-10|=10 > |3|=3$.

18. Prime numbers

DEFINITION - Formal definition

A positive integer $p > 1$ is prime when its only positive factors are 1 and p . The number 1 is not prime. A number greater than 1 that is not prime is composite.

To test n for primality, it is enough to test prime divisors not exceeding $\sqrt{n}$. If $n = ab$ with $1 < a \leq b$, then $a \leq \sqrt{n}$; therefore any non-trivial factorisation reveals a smaller factor at or below $\sqrt{n}$.

MEDIUM WORKED EXAMPLE - Test 221

Problem. Determine whether 221 is prime.

Solution.

$\sqrt{221}$ is between 14 and 15, so test primes 2,3,5,7,11,13. The number is not divisible by 2,3,5,7 or 11, but $221 = 13 \cdot 17$. Therefore 221 is composite.

REALLY HARD WORKED EXAMPLE - Test 1009 efficiently

Problem. Determine whether 1009 is prime, showing a complete argument.

Solution.

$\sqrt{1009} \approx 31.8$, so test prime divisors 2,3,5,7,11,13,17,19,23,29,31. It is odd; digit sum 10 excludes 3; last digit excludes 5. Direct division gives non-zero remainders for 7,11,13,17,19,23,29,31. Since no prime $\leq \sqrt{1009}$ divides it, 1009 is prime.

19. Factorial notation

DEFINITION - Definition

For a positive integer n ,

$$n! = n(n-1)(n-2)\dots 3\cdot 2\cdot 1.$$

Also, standard mathematics defines $0!=1$, although Unit 1 states the definition beginning with positive n .

The recursive form is $n!=n(n-1)!$. Factorials grow rapidly and appear later in counting, permutations, combinations and algorithm analysis.

MEDIUM WORKED EXAMPLE - Expand and evaluate

Problem. Evaluate $7!$ and relate it to $6!$.

Solution.

$7!=7\cdot 6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1=5040$. Since $6!=720$, the recursive relation gives $7!=7\cdot 6!=7\cdot 720=5040$.

REALLY HARD WORKED EXAMPLE - Cancel factorial structure

Problem. Evaluate $12!/(10!\cdot 2!)$ without expanding either factorial completely.

Solution.

$12!=12\cdot 11\cdot 10!$, so

$$12!/(10!\cdot 2!)=(12\cdot 11\cdot 10!)/(10!\cdot 2\cdot 1)=(12\cdot 11)/2=66.$$

Cancel only common non-zero factors; expanding $12!$ fully creates unnecessary arithmetic.

20. Law 5: linearity (trichotomy)

DEFINITION - Formal law

For all $m, n \in \mathbb{Z} \geq 0$, exactly one of these is true: $m < n$, $m = n$, $m > n$. The same trichotomy holds in $\mathbb{Z}$. 'Exactly one' means the cases are mutually exclusive and collectively exhaustive.

Linearity ensures any two integers are comparable. It is stronger than merely saying at least one relation holds, because no two of the three can be true simultaneously.

MEDIUM WORKED EXAMPLE - Identify the unique case

Problem. Let $m = 3 \cdot 14$ and $n = 6 \cdot 7$. Which trichotomy case holds?

Solution.

$m = 42$ and $n = 42$, so $m = n$. Consequently $m < n$ and $m > n$ are both false.

REALLY HARD WORKED EXAMPLE - Parameter-dependent trichotomy

Problem. For $k \in \mathbb{Z}$, compare $m = 2k + 7$ and $n = 5k - 2$ in all possible cases.

Solution.

Compute $n - m = 3k - 9 = 3(k - 3)$. If $k > 3$, $n - m > 0$, so $m < n$. If $k = 3$, $m = n$. If $k < 3$, $n - m < 0$, so $m > n$. These cases are exhaustive and mutually exclusive by trichotomy.

21. Law 6: monotonicity of addition and multiplication

DEFINITION - Formal law

If $m=n$, then $m+k=n+k$ and $mk=nk$.

If $m<n$, then $m+k<n+k$.

If $m<n$ and $k>0$, then $mk<nk$.

For $\mathbb{Z}$, if $m<n$ and $k<0$, then $mk>nk$: a negative multiplier reverses the inequality.

Adding the same integer preserves order. Multiplication preserves order only for a positive multiplier; zero collapses both sides to equality; a negative multiplier reverses order.

MEDIUM WORKED EXAMPLE - Apply monotonicity

Problem. From $-4<7$, determine the results after (a) adding -9 and (b) multiplying by -3 .

Solution.

(a) $-4-9<7-9$, so $-13<-2$. (b) Since $-3<0$, reverse the sign: $(-4)(-3)>(7)(-3)$, so $12>-21$.

REALLY HARD WORKED EXAMPLE - Solve an integer inequality

Problem. Find every integer x satisfying $-5(2x-3)\geq 35$.

Solution.

Expand: $-10x+15\geq 35$. Add -15 : $-10x\geq 20$. Dividing by -10 reverses the inequality, giving $x\leq -2$. Therefore the integer solution set is $\{\dots, -5, -4, -3, -2\}$. Check the boundary: $x=-2$ gives $35\geq 35$.

COMMON TRAP - Zero multiplier

From $m<n$ you do not get $m\cdot 0<n\cdot 0$; both sides become 0 . That is why the strict multiplication clause requires $k>0$ or $k<0$, not $k\geq 0$.

22. Law 7: transitivity of equality and order

DEFINITION - Formal law

If $m=n$ and $n=k$, then $m=k$.

If $m<n$ and $n<k$, then $m<k$.

Transitivity allows a chain to pass through a shared middle value.

The middle terms must connect correctly. From $a<b$ and $c<d$ alone, nothing follows about a and d . Also, 'not equal' is not transitive: $1\neq 2$ and $2\neq 1$, but $1=1$.

MEDIUM WORKED EXAMPLE - Build an inequality chain

Problem. Given $a+4<b+4$ and $b-9<c-9$, show $a<c$.

Solution.

By order preservation under equal addition, $a<b$ and $b<c$. By transitivity of $<$, $a<c$.

REALLY HARD WORKED EXAMPLE - Prove a chained result with mixed laws

Problem. Suppose $a<b<c$ and $k<0$. Prove $ak>ck$.

Solution.

From $a<b$ and $k<0$, monotonicity gives $ak>bk$. From $b<c$ and $k<0$, $bk>ck$. Now apply transitivity of $>$: $ak>ck$. The reversal occurs at each multiplication step; transitivity joins the results.

23. The complete law system for $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$

The guide concludes with nine laws for $\mathbb{Z}_{\geq 0}$. All nine also hold for $\mathbb{Z}$, but the multiplication part of monotonicity requires the additional negative-multiplier case. $\mathbb{Z}$ also has a tenth law: every integer has an additive inverse within $\mathbb{Z}$.

LAW	NAME	CORE STATEMENT
1	Commutativity	$m+n=n+m$; $mn=nm$
2	Associativity	$m+(n+k)=(m+n)+k$; $m(nk)=(mn)k$
3	Distributivity	$m(n+k)=mn+mk$
4	Multiplicative identity	$m \cdot 1 = m$
5	Linearity	exactly one: $m < n$, $m = n$, $m > n$
6	Monotonicity	same addition preserves order; positive product preserves; negative product reverses
7	Transitivity	if $m=n$ and $n=k$, then $m=k$; if $m < n$ and $n < k$, then $m < k$
8	Additive identity	$m+0=m$
9	No zero-divisors	$mn=0$ iff $m=0$ or $n=0$
10 ($\mathbb{Z}$ only)	Additive inverses	for every m , some n satisfies $m+n=0$

MEDIUM WORKED EXAMPLE - Audit a calculation

Problem. Explain which laws justify $6(23+77)=600$.

Solution.

Distributivity gives $6 \cdot 23 + 6 \cdot 77$. Equivalently, evaluate inside first: $23+77=100$, then multiply to get 600. Associativity and ordinary closure keep all intermediate values in $\mathbb{Z}_{\geq 0}$. No additive inverse is required.

REALLY HARD WORKED EXAMPLE - Compare the three systems by laws

Problem. For each system $\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$, decide whether it contains an additive identity and whether every element has an additive inverse within the same system.

Solution.

$\mathbb{Z}^+$: no additive identity because $0 \notin \mathbb{Z}^+$; no element's inverse lies in $\mathbb{Z}^+$.

$\mathbb{Z}_{\geq 0}$: contains additive identity 0; only 0 has its additive inverse in the system, so the universal inverse law fails.

$\mathbb{Z}$: contains additive identity 0, and every x has $-x \in \mathbb{Z}$, so the universal inverse law holds.

NOTE - Law 7 wording

Write equality transitivity as: if $m=n$ and $n=k$, then $m=k$. A compressed chain such as ' $m=n=k$ ' is readable, but the conditional form is the formal law.

24. Master comparison matrix

Use this table to separate set membership, closure and special elements. A tick means the property holds throughout that system.

Feature	$\mathbb{Z}^+$	$\mathbb{Z}_{\geq 0}$	$\mathbb{Z}$
Contains 0	No	Yes	Yes
Contains negative integers	No	No	Yes
Closed under +	Yes	Yes	Yes
Closed under $\times$	Yes	Yes	Yes
Closed under -	No	No	Yes
Additive identity present	No	Yes	Yes
Multiplicative identity present	Yes	Yes	Yes
Every element has additive inverse inside	No	No	Yes
No zero-divisors	Not applicable: 0 absent	Yes	Yes
Linearly ordered	Yes	Yes	Yes

MEDIUM WORKED EXAMPLE - Use the matrix

Problem. Why can $x+7=3$ be solved in $\mathbb{Z}$ but not in $\mathbb{Z}_{\geq 0}$?

Solution.

The algebra gives $x=-4$. This is an integer, so it is valid in $\mathbb{Z}$. It is not non-negative, so no solution exists within $\mathbb{Z}_{\geq 0}$. The equation requires the additive inverse of 4, which lies outside $\mathbb{Z}_{\geq 0}$.

REALLY HARD WORKED EXAMPLE - Determine structural failure

Problem. A student claims $\mathbb{Z}_{\geq 0}$ behaves exactly like $\mathbb{Z}$ because both share Laws 1-9. Give the sharpest correction.

Solution.

Sharing Laws 1-9 does not make the systems identical. $\mathbb{Z}$ has negative elements and satisfies Law 10: every element has an additive inverse in the system. $\mathbb{Z}_{\geq 0}$ lacks inverses for every positive member. Its multiplication monotonicity also never encounters a negative multiplier, whereas the $\mathbb{Z}$ version must include inequality reversal for $k < 0$.

25. Exam traps checklist

COMMON TRAP - Check

Do not include 0 in $\mathbb{Z}^+$ under the guide's notation.

COMMON TRAP - Check

Do not say $-x$ must be negative; its sign depends on x .

COMMON TRAP - Check

Do not confuse commutativity (order) with associativity (grouping).

COMMON TRAP - Check

Do not claim addition distributes over multiplication.

COMMON TRAP - Check

Do not apply the zero-product property unless the product equals 0.

COMMON TRAP - Check

Do not write $\sqrt{(x^2)}=x$ without knowing $x \geq 0$; write $|x|$.

COMMON TRAP - Check

Do not forget the second branch when equal squares are solved.

COMMON TRAP - Check

Do not reverse an inequality when adding; reverse it when multiplying or dividing by a negative number.

COMMON TRAP - Check

Do not call 1 prime.

COMMON TRAP - Check

Do not use an illegal digit for a chosen base.

COMMON TRAP - Check

Do not treat several examples as proof of a universal law.

26. Final mixed challenge

Attempt these without looking back. Full solutions follow immediately so you can mark reasoning, not only final answers.

1. Determine the smallest system among $\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}$ that contains every value of $(a-b)(a+b)$ for $a, b \in \mathbb{Z}^+$.
2. Factor $30x^5 - 75x^3$ completely using distributivity.
3. Convert $(24031)_5$ to decimal.
4. Solve $(x^2 - 16)(x^2 - 25) = 0$ over $\mathbb{Z}$.
5. Solve $|2x - 3| = 11$ over $\mathbb{Z}$.
6. Determine whether 391 is prime.
7. Evaluate $11!/(8! \cdot 3!)$.
8. If $-4 < a < 3$ and $k = -5$, state a correct inequality involving ak .
9. Identify the exact error in: $(-3)^2 = 3^2$, therefore $-3 = 3$.
10. Name the law that $\mathbb{Z}$ has but $\mathbb{Z}_{\geq 0}$ lacks.

Worked solutions

1. $(a-b)(a+b) = a^2 - b^2$ can be positive, zero or negative; for example (5,2), (3,3), (2,5). The smallest listed system is $\mathbb{Z}$.
2. $30x^5 - 75x^3 = 15x^3(2x^2 - 5)$. Over integer coefficients, the remaining quadratic does not factor further.
3. $(24031)_5 = 2 \cdot 5^4 + 4 \cdot 5^3 + 0 \cdot 5^2 + 3 \cdot 5 + 1 = 1250 + 500 + 15 + 1 = 1766$.
4. Zero product: $x^2 = 16$ or $x^2 = 25$. Thus $x \in \{-5, -4, 4, 5\}$.
5. $2x - 3 = 11$ gives $x = 7$; $2x - 3 = -11$ gives $x = -4$. Both are integers.
6. $391 = 17 \cdot 23$, so it is composite.
7. $11!/(8! \cdot 3!) = (11 \cdot 10 \cdot 9)/(3 \cdot 2 \cdot 1) = 165$.
8. From $-4 < a$, multiplying by -5 gives $20 > ak$. From $a < 3$, multiplying by -5 gives $ak > -15$. Therefore $-15 < ak < 20$.
9. From equal squares, one may conclude equal absolute values, not equal signed bases:
 $\sqrt{((-3)^2)} = |-3| = 3$ and $\sqrt{(3^2)} = |3| = 3$.
10. Law 10, existence of additive inverses for every element: for each $m \in \mathbb{Z}$, there exists $n \in \mathbb{Z}$ with $m + n = 0$.

27. One-page revision sheet

CONCEPT	MUST-KNOW FORM
System	$\mathbb{Z}^+ = \{1, 2, \dots\}$; $\mathbb{Z}_{\geq 0} = \{0, 1, 2, \dots\}$; $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
Containment	$\mathbb{Z}^+ \subset \mathbb{Z}_{\geq 0} \subset \mathbb{Z}$
Commutative	$a+b=b+a$; $ab=ba$
Associative	$a+(b+c)=(a+b)+c$; $a(bc)=(ab)c$
Distributive	$a(b+c)=ab+ac$
Identities	$a+0=a$; $a \cdot 1=a$
Zero product	$ab=0$ iff $a=0$ or $b=0$
Additive inverse	$a+(-a)=0$; $-a$ need not be negative
Square root	$\sqrt{a^2}= a $
Order	$a < b$ iff $b-a \in \mathbb{Z}^+$
Monotonicity	add same: preserve; $\times$ positive: preserve; $\times$ negative: reverse
Absolute value	distance from 0; $ a-b $ is distance between a and b
Prime	$p > 1$ with positive factors only 1, p
Factorial	$n! = n(n-1)\dots 2 \cdot 1$; $0! = 1$
Extra $\mathbb{Z}$ law	every integer has an additive inverse in $\mathbb{Z}$

NOTE - Source alignment

Coverage is based on Study Unit 1 (pages 1-20) of the supplied COS1501 study guide. The explanations and worked examples here are newly written and extend the guide's difficulty while staying within its concepts.

EXAM NOTE - Readiness standard

You are ready to move on when you can state each law formally, identify its domain, name the law used at each algebraic step, reject invalid square-root and zero-product reasoning, and solve the hard examples without copying their method.